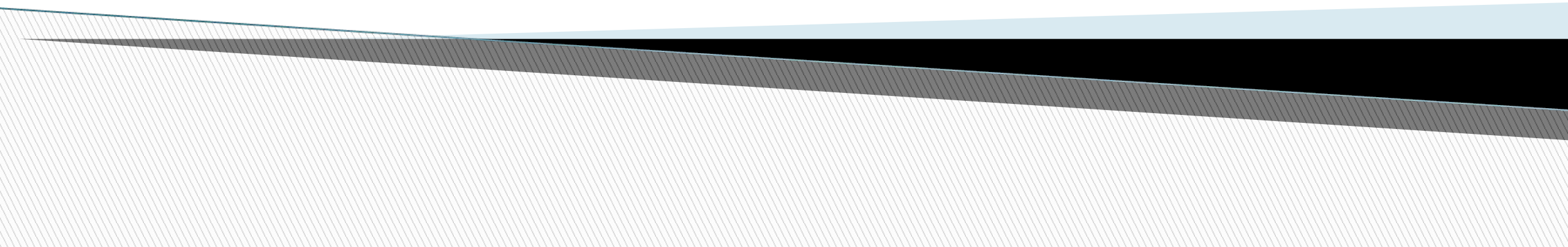
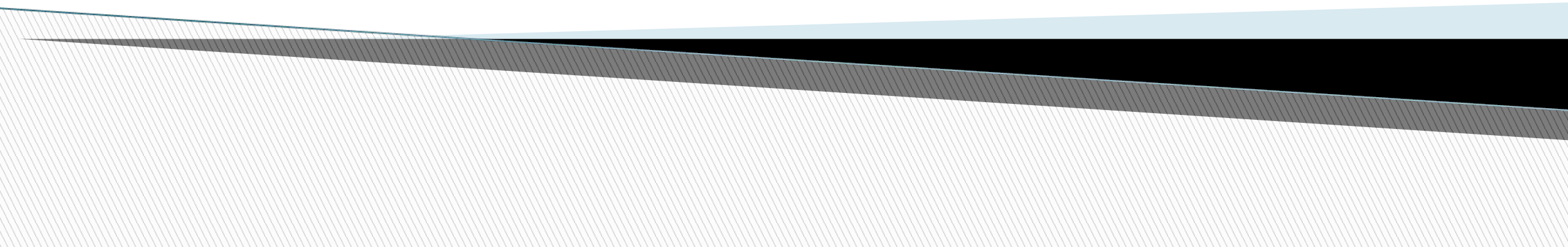


# Database System

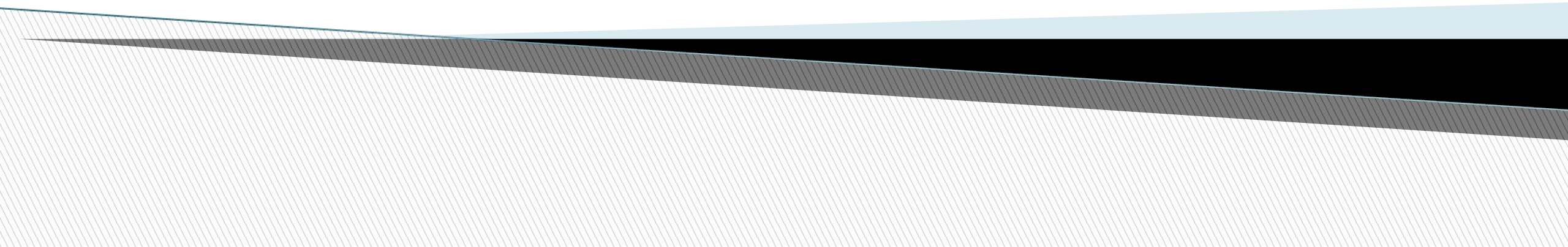
ER-Data Model



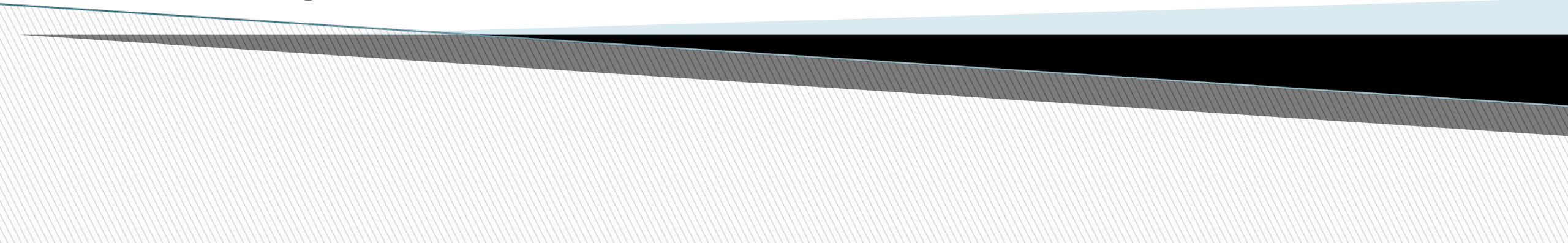
# ER Data Model

- ◆ The Entity-Relationship (E-R) model is a popular conceptual data modeling technique used to represent and visualize relationships between entities in a database system.
  - ◆ The E-R model employs a graphical approach to illustrate the structure of a database, emphasizing entities, their attributes, and the relationships among them.
- 
- A decorative graphic at the bottom of the slide consisting of several overlapping, wavy horizontal bands. The colors include light blue, dark blue, and grey, with some areas having a fine diagonal line pattern.

# How it differs from relational data model

- ❖ The Relational model is a practical implementation that uses tables for data storage, whereas the E-R model is a **high-level conceptual tool for visualizing entities and relationships**.
  - ❖ Relational offers the framework for database systems, combining conceptual clarity with effective implementation, whereas E-R helps with design.
- 

# Requirements Become the E-R Data Model

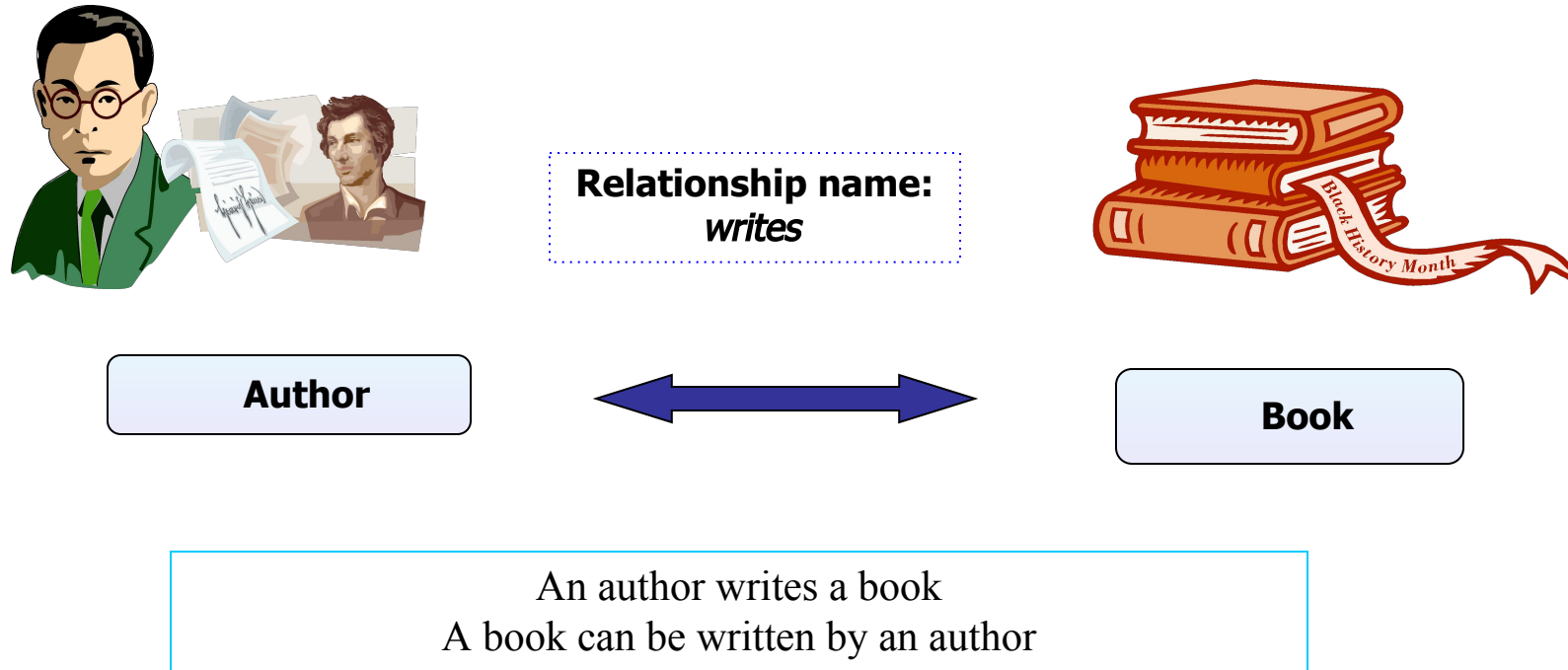
- After the requirements have been gathered, they are transformed into an Entity Relationship (E-R) Data Model
  - E-R Models consist of
    - Entities
    - Attributes
    - Identifiers
    - Relationships
- 
- The bottom of the slide features a decorative graphic consisting of several overlapping, wavy horizontal bands. From top to bottom, the bands are light blue, black, dark grey, and white with a fine diagonal line pattern.

# E-R Diagram

- ❖ A picture showing the information created, stored, and used by a business system.
- ❖ ERD is network model that describes stored data of a system at a high level of abstraction.
- ❖ Used to design database applications
- ❖ Tool for communication between database designers and users
- ❖ Entities generally represent similar kinds of information
- ❖ Lines drawn between entities show relationships among the data
- ❖ High level business rules are also shown

# Relationship

- ? A relationship is an association among several entities
  - Relationship name is an active or a passive verb.

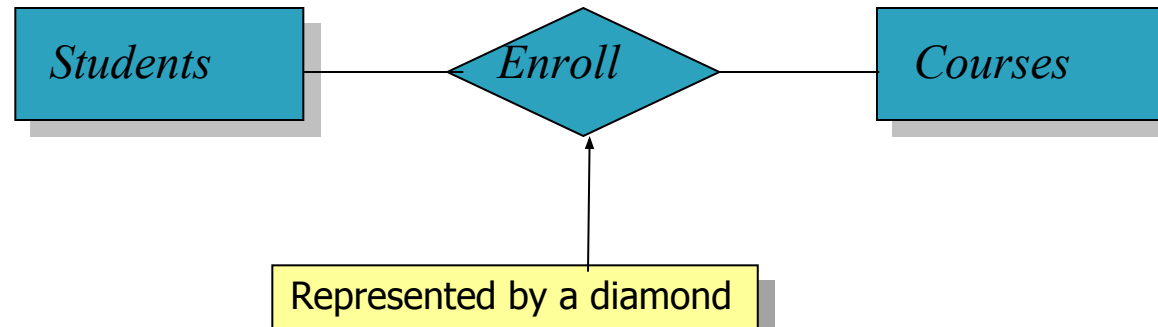


# Relationship

- ? Given a *customer* and an *account*, the relationship *deposits* between them indicates that the customer deposits money into the account

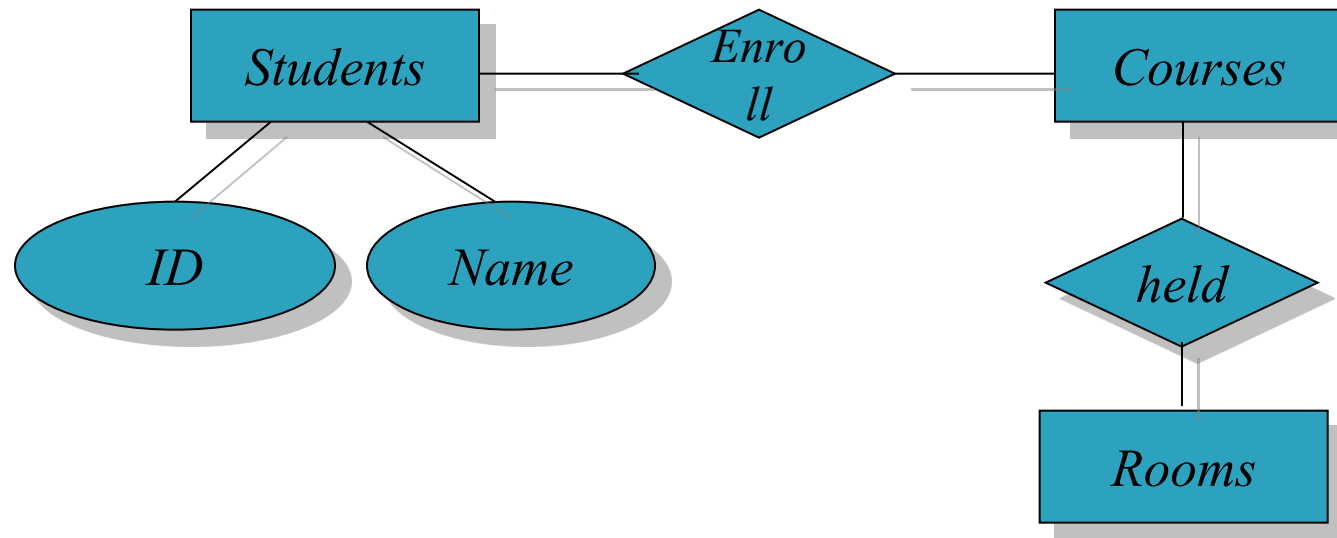
|                        |                 |                       |
|------------------------|-----------------|-----------------------|
| <u>Hayes</u>           | <u>deposits</u> | <u>A-102</u>          |
| <i>customer</i> entity | relationship    | <i>account</i> entity |

- ? Students enroll in courses



# Relationships

- ? Students *enroll* in courses
- ? Courses are *held* in rooms





# Relationship Sets

- ? A relationship set is a set of relationships of the same type
- ? *Mathematically, a relationship set* is a relation among  $n \geq 2$  entity sets. If  $E_1, E_2, \dots, E_n$  are entity sets then a relationship set R is a subset of:

$$\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$$

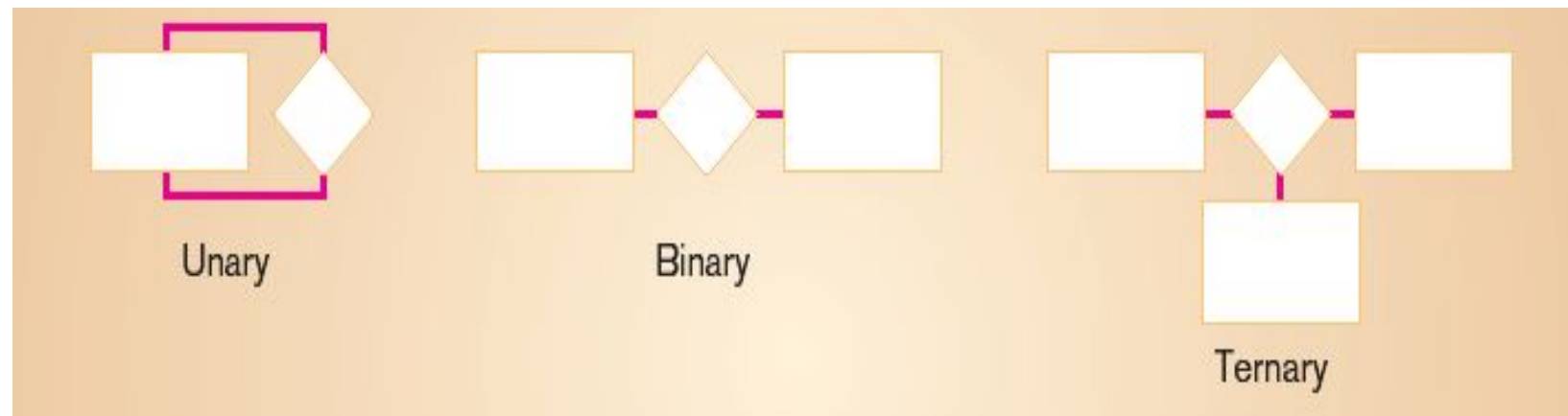
where  $(e_1, e_2, \dots, e_n)$  is a relationship

- Example:

$$(\text{Hayes}, \text{A-102}) \in \textit{Deposits}$$

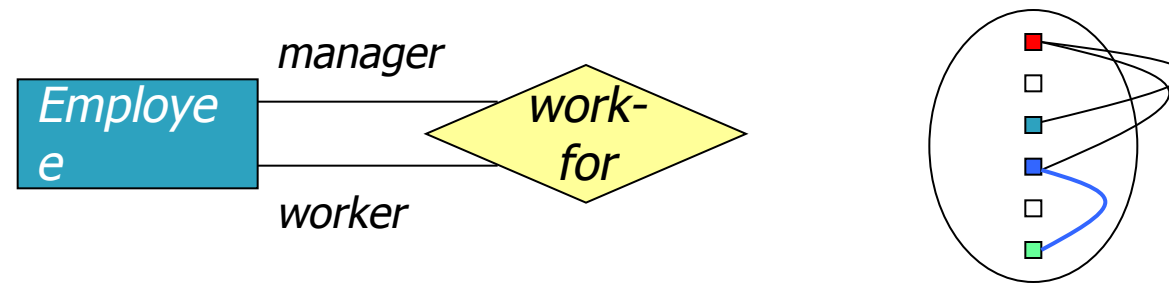
# Degree of Relationships

- ? Degree: number of entity types that participate in a relationship
- ? Three cases
  - **Unary:** between two instances of one entity type
  - **Binary:** between the instances of two entity types
  - **Ternary:** among the instances of three entity types

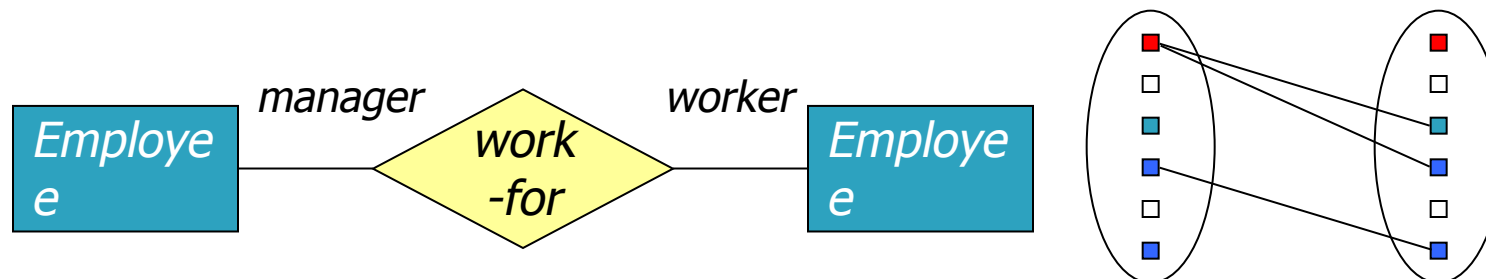


# Unary/Recursive Relationships

- A relationship relating entitles of the same type



- Employees play different roles: **manager** or **worker**
- A recursive relationship to understand it:



# Unary/Recursive Relationships

- Without Role Names

| <i>EmployeeNo</i> | <i>EmployeeNo</i> |
|-------------------|-------------------|
| <i>A1234</i>      | <i>A6543</i>      |
| <i>A1234</i>      | <i>A8734</i>      |

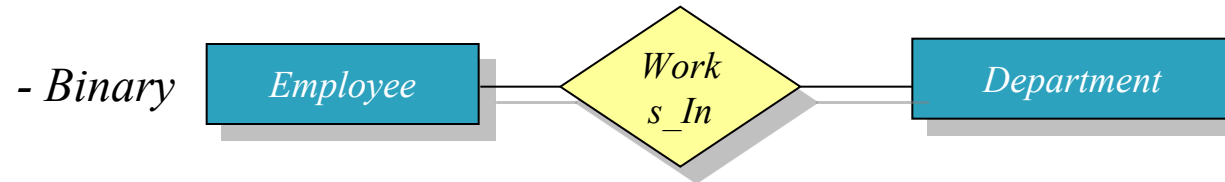
- With Role Names

| <i>ManagerNo</i> | <i>WorkerNo</i> |
|------------------|-----------------|
| <i>A1234</i>     | <i>A6543</i>    |
| <i>A1234</i>     | <i>A8734</i>    |

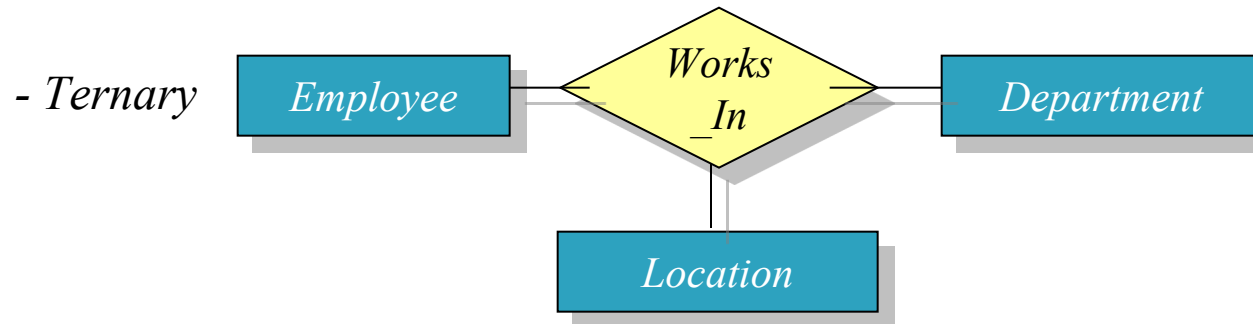
Where ManagerNo and WorkerNo are Valid EmployeeNo

# Binary & Ternary Relationships

? *Employee* *works* in a *Department*

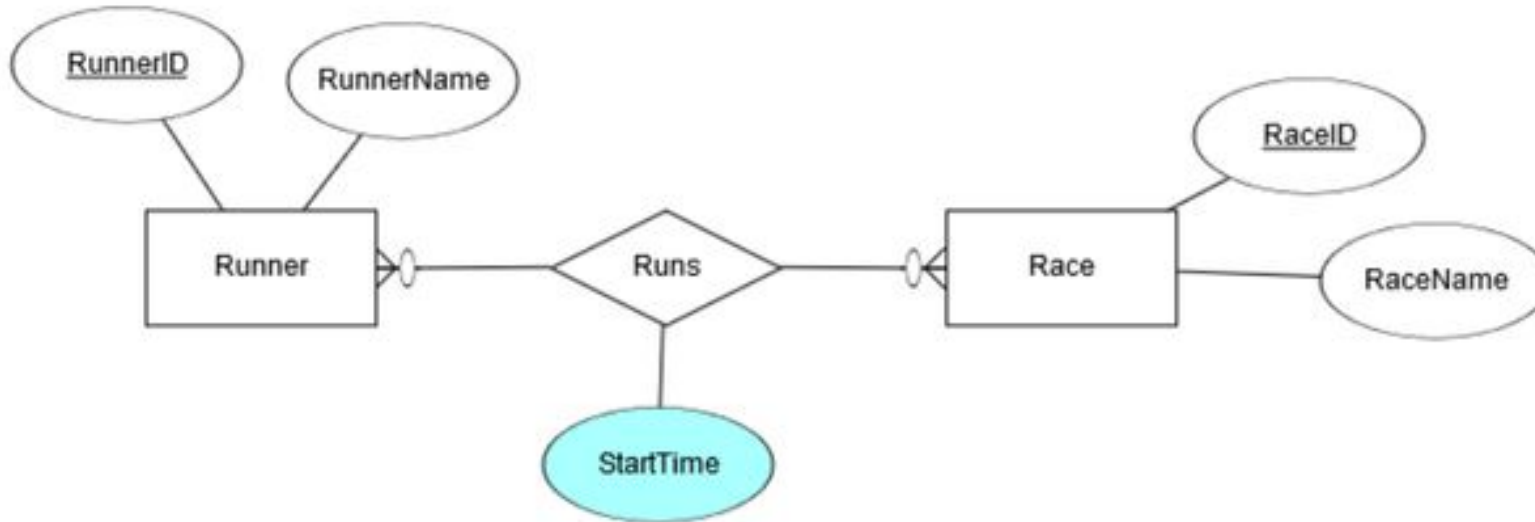


■ *Employee* *works* in a *Department* at a particular *location*



# Associating Attributes with Relationship

- ? In many to many relationship, attributes may also be associated with the relationship.

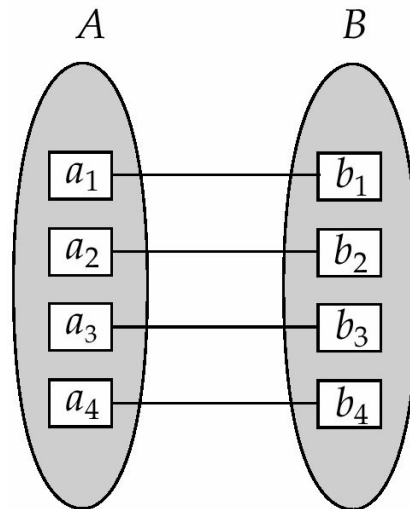


# Mapping Cardinalities

- ? Express the number of entities to which another entity can be associated via a relationship set
- ? Most useful in describing binary relationship sets
- ? For a binary relationship set the mapping cardinality must be one of the following types:
  - One to one
  - One to many
  - Many to one
  - Many to many

# One to One (1:1)

An entity in A is associated with *at most* one entity in B, and an entity in B is associated with *at most* one entity in A



Some elements in A and B may not be mapped to any elements in the other set

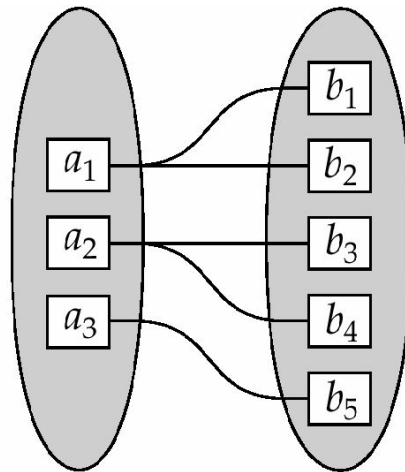


A customer can borrow 1 loan and vice versa



# One to Many (1:M)

An entity in A is associated with *any number* (zero or more) of entities in B. An entity in B, however, can be associated with *at most* one entity in A



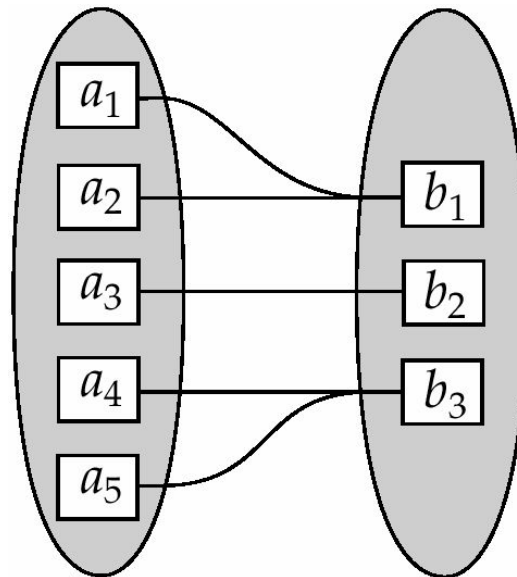
Some elements in A and B may not be mapped to any elements in the other set



A Customer can borrow more than 1 loan, whereas a loan has only one borrower

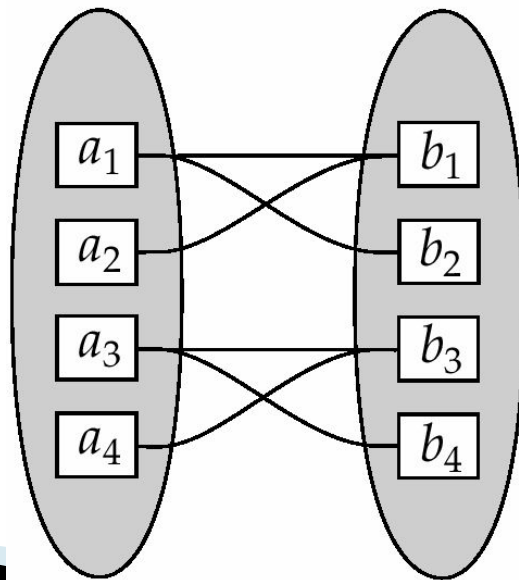
## Many to One (M: 1)

An entity in A is associated with *at most* one entity in B. An entity in B, however, can be associated with *any number* (zero or more) of entities in A



# Many to Many (M:N)

An entity in A is associated with *any number* (zero or more) of entities in B and an entity in B is associated with *any number* (zero or more) of entities in A



- A customer can borrow more than one loan
- A loan can have more than one borrower.

# Participation Constraints

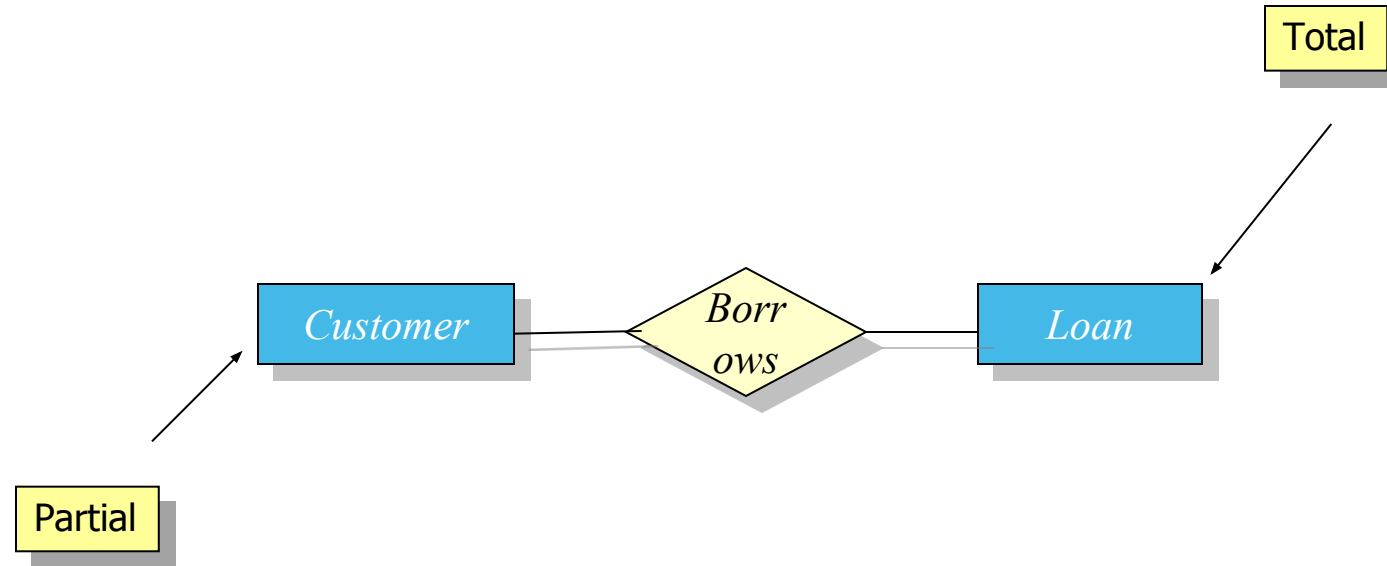
## Total Participation

The participation of an entity set  $E$  in a relationship set  $R$  is said to be *total* if every entity in  $E$  participates in at least one relationship in  $R$

## Partial Participation

If only some entities in  $E$  participate in relationships in  $R$ , the participation of entity set  $E$  in relationship  $R$  is said to be *partial*

# Participation Constraints

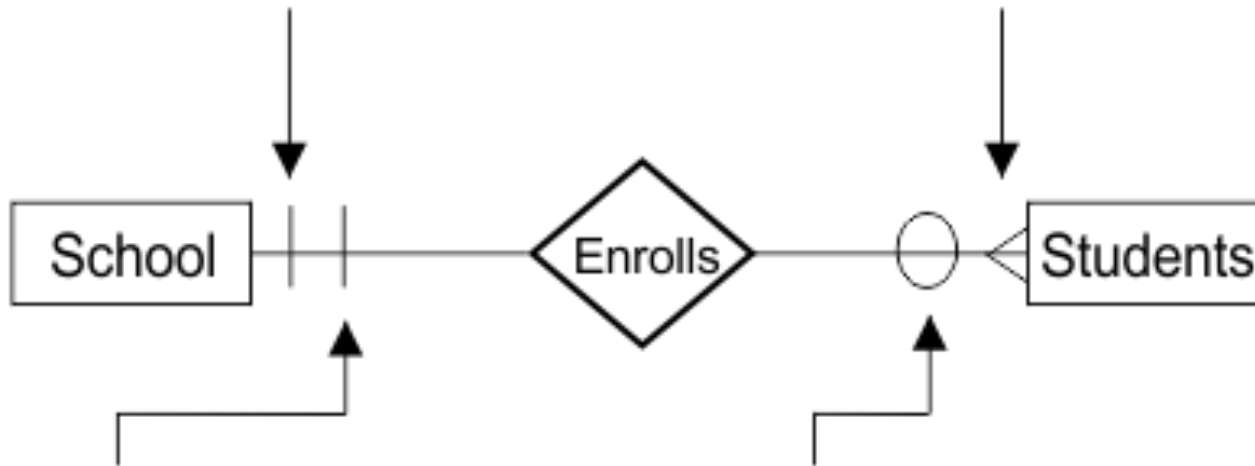


Every loan must have a customer  
Every customer (of the bank) does not borrow a loan

# Modality of Relationship

## Cardinality

Indicates that **only one** school is involved in the relationship



## Modality: Mandatory

Indicates that **there must be one** school to have this relationship

## Cardinality

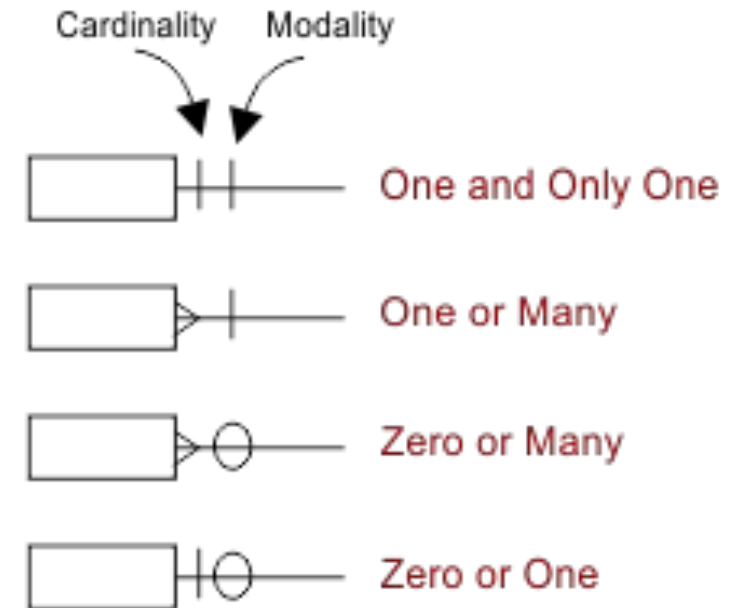
Indicates that there may be **Many** students in Schools

## Modality: Optional

Indicates that there may be **No** students in Schools

## Notations

- | Represent one
- Represent Zero
- ← Represent Many



# Identify the Relationships

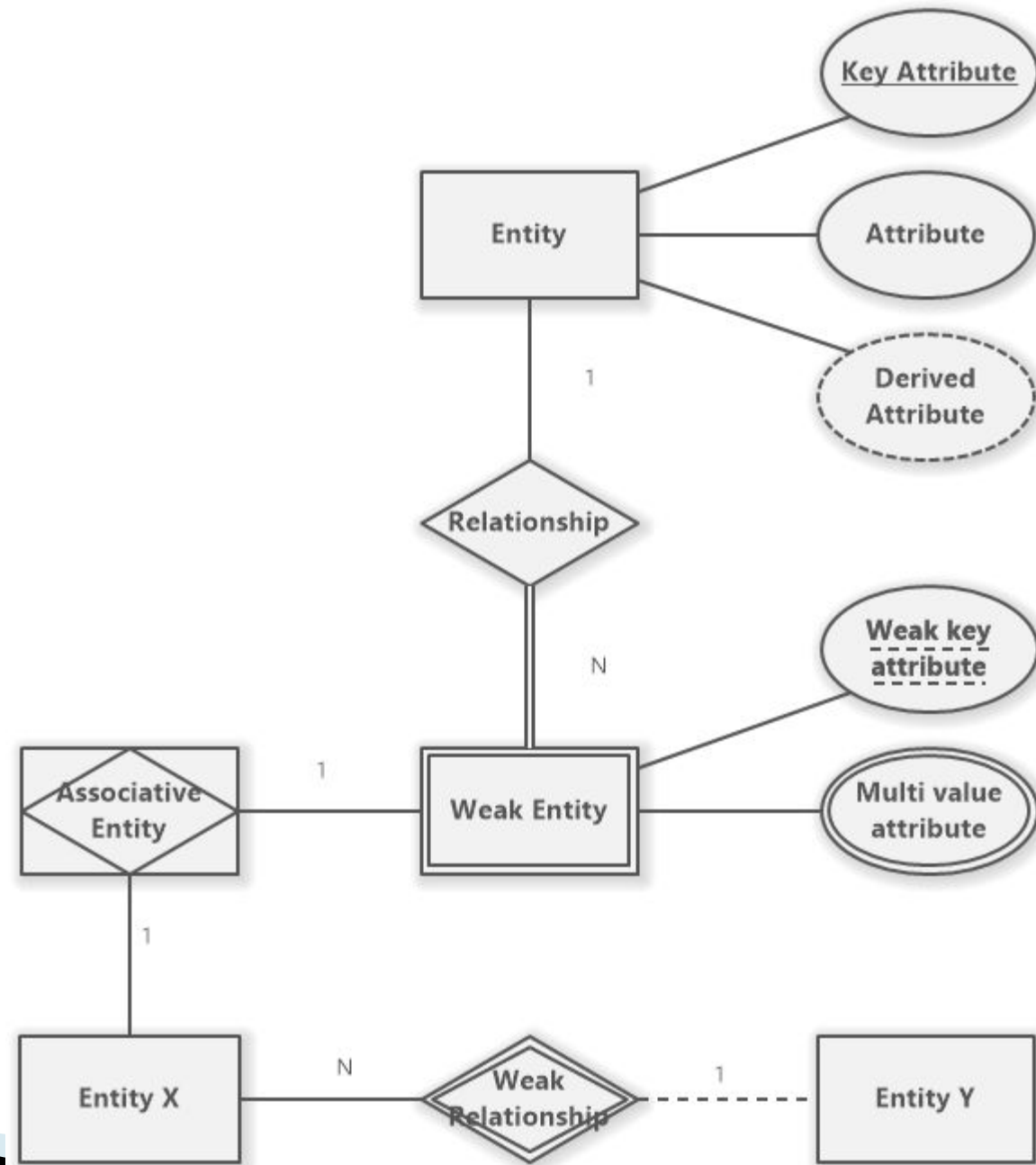
- ? One department offers many courses. But one particular course can be offered by only one department. hence the cardinality between department and course is One to Many (1:N)
- ? One department has multiple instructors . But instructor belongs to only one department. Hence the cardinality between department and instructor is One to Many (1:N)
- ? One department has only one head and one head can be the head of only one department. Hence the cardinality is one to one. (1:1)
- ? One course can be enrolled by many students and one student can enroll for many courses. Hence the cardinality between course and student is Many to Many (M:N)
- ? One course is taught by only one instructor. But one instructor teaches many courses. Hence the cardinality between course and instructor is Many to One (N :1)

# Degree of Relationship

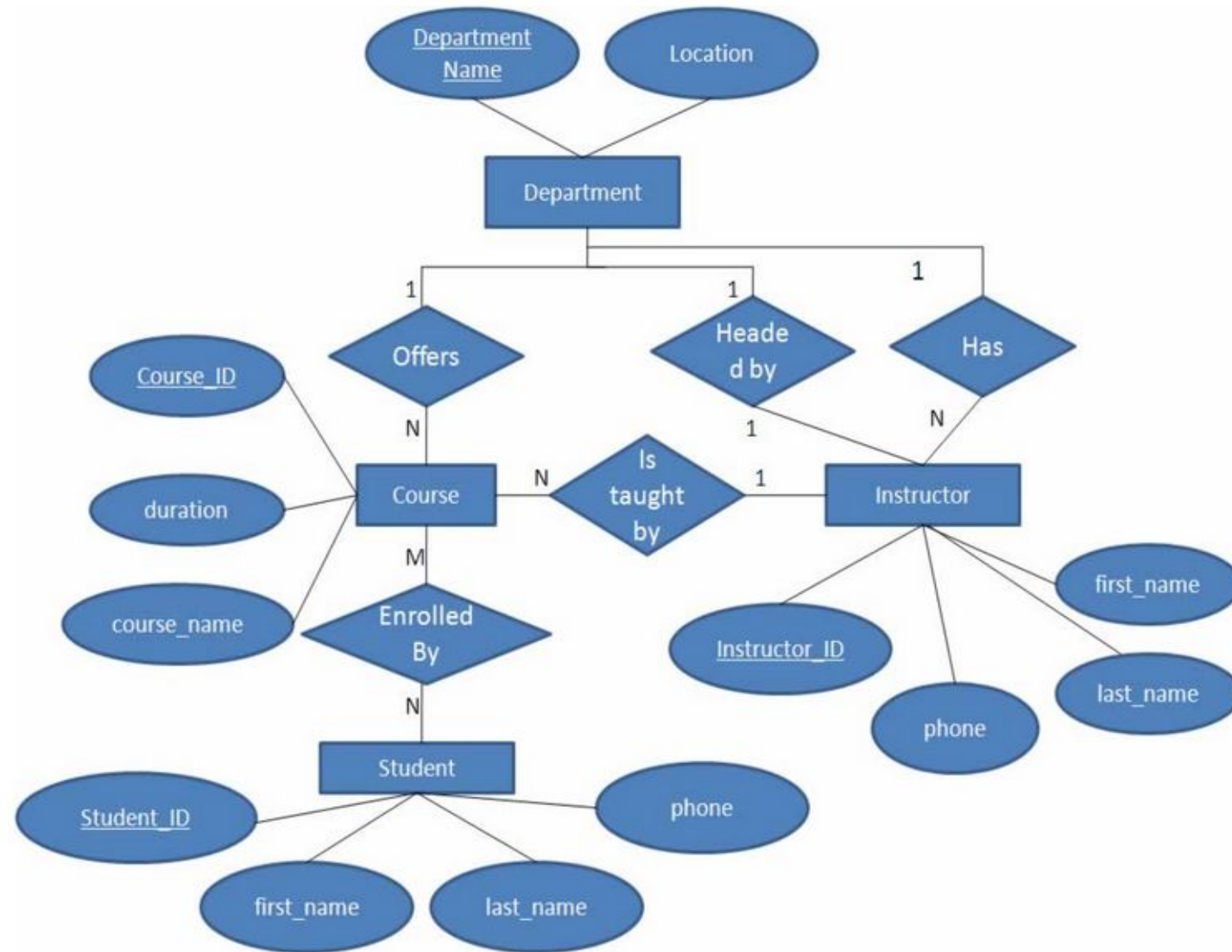
## ? Between

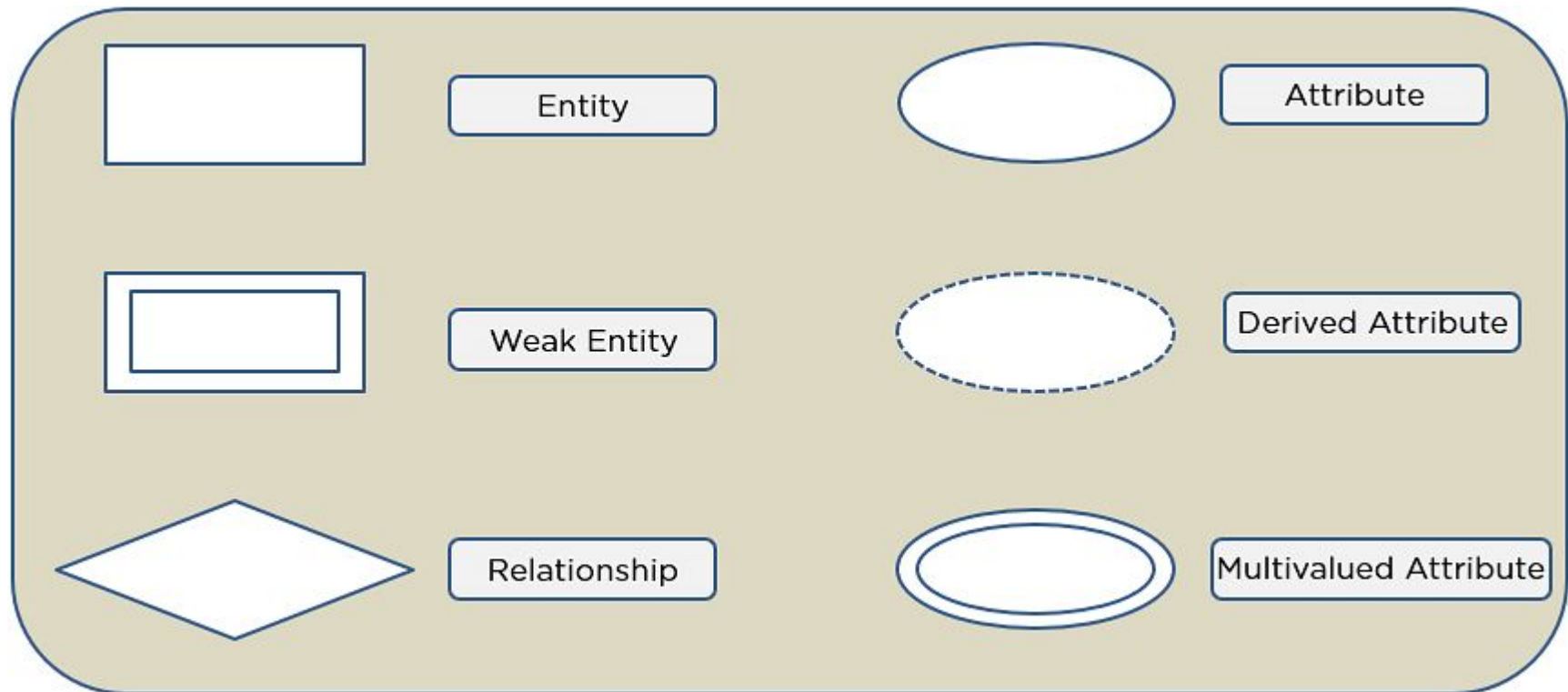
- department and course
- department and instructor
- course and student
- course and instructor



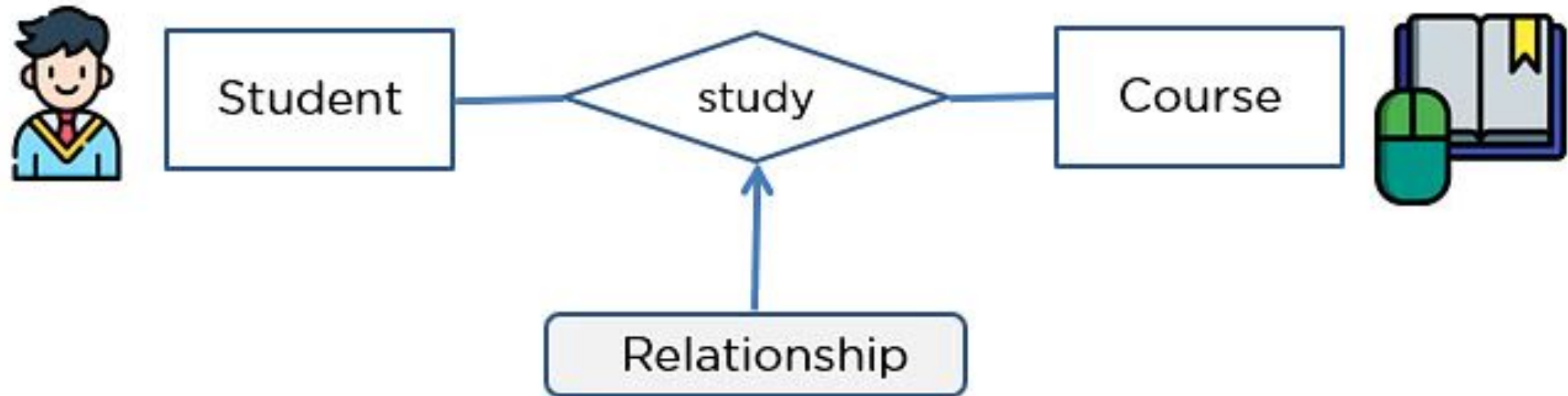


# Complete E-R Diagram



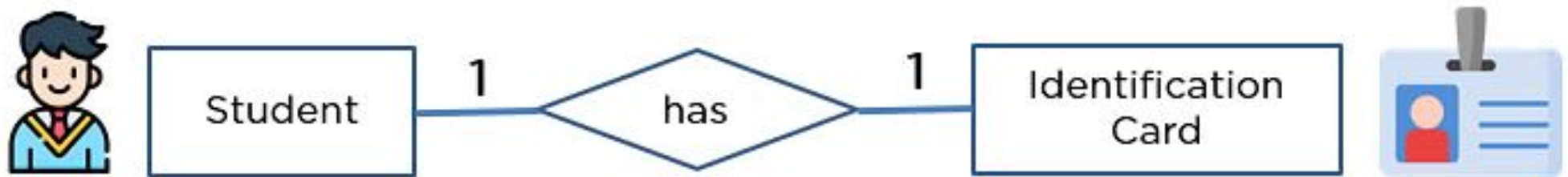


# Relationship



## One-to-One Relationship

For example, a student has only one identification card and an identification card is given to one person.



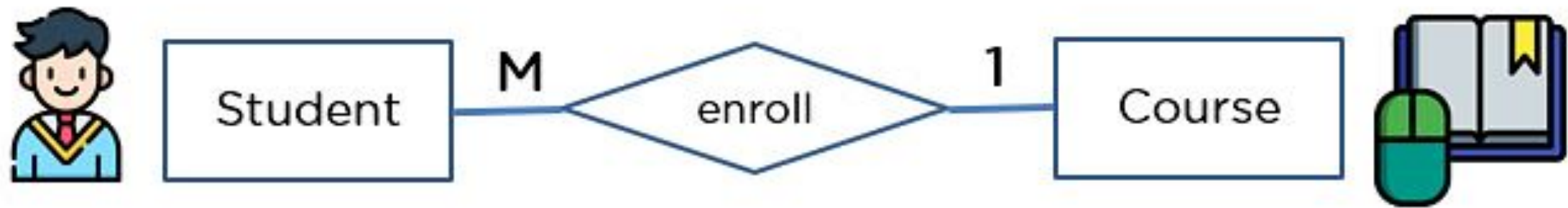
## One-to-Many Relationship

For example, a customer can place many orders, but an order cannot be placed by many customers



## Many-to-One Relationship

For example, students have to opt for a single course, but a course can have many students



## Many-to-Many Relationship

For example, you can assign an employee to many projects and a project can have many employees.

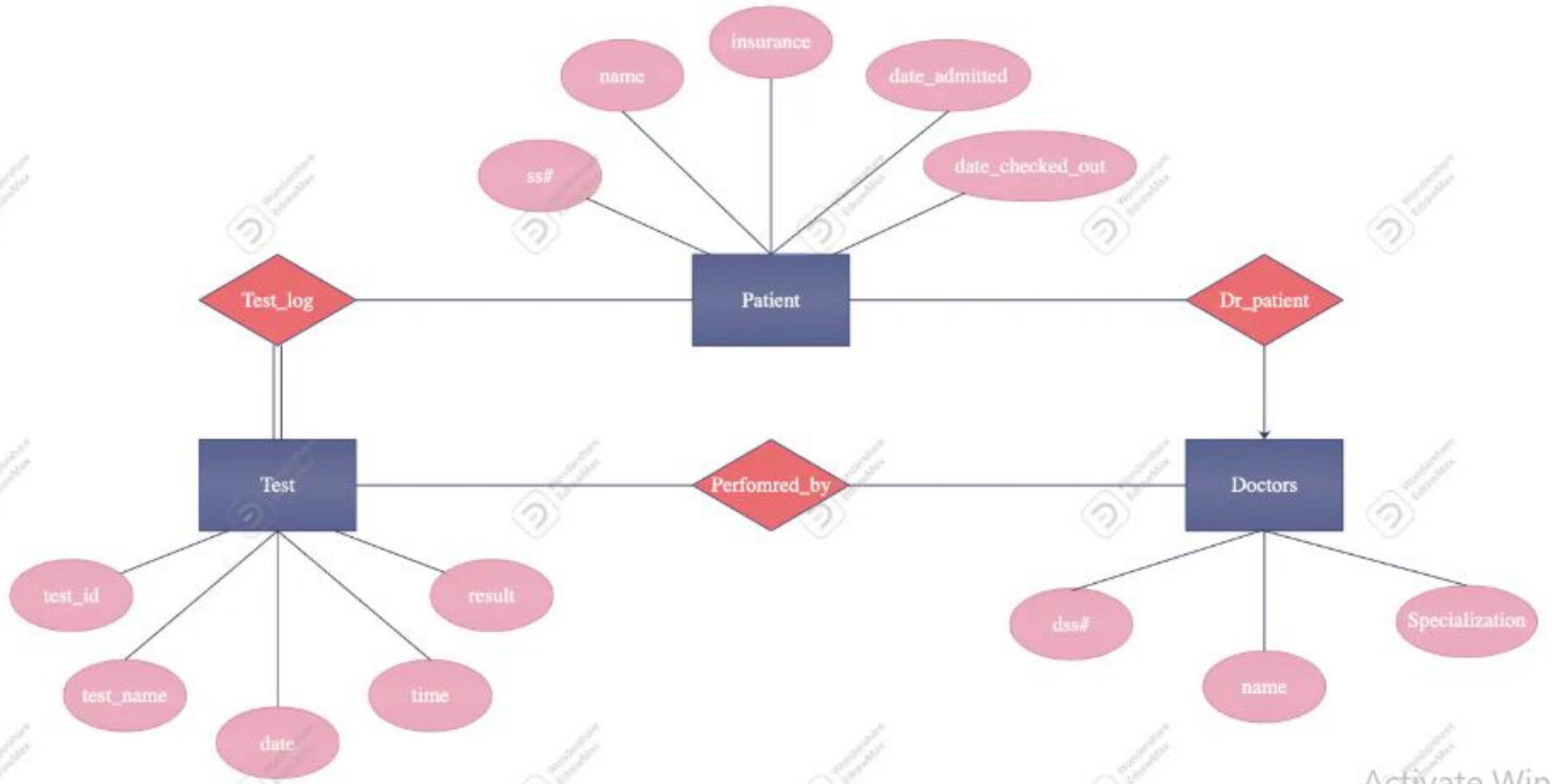




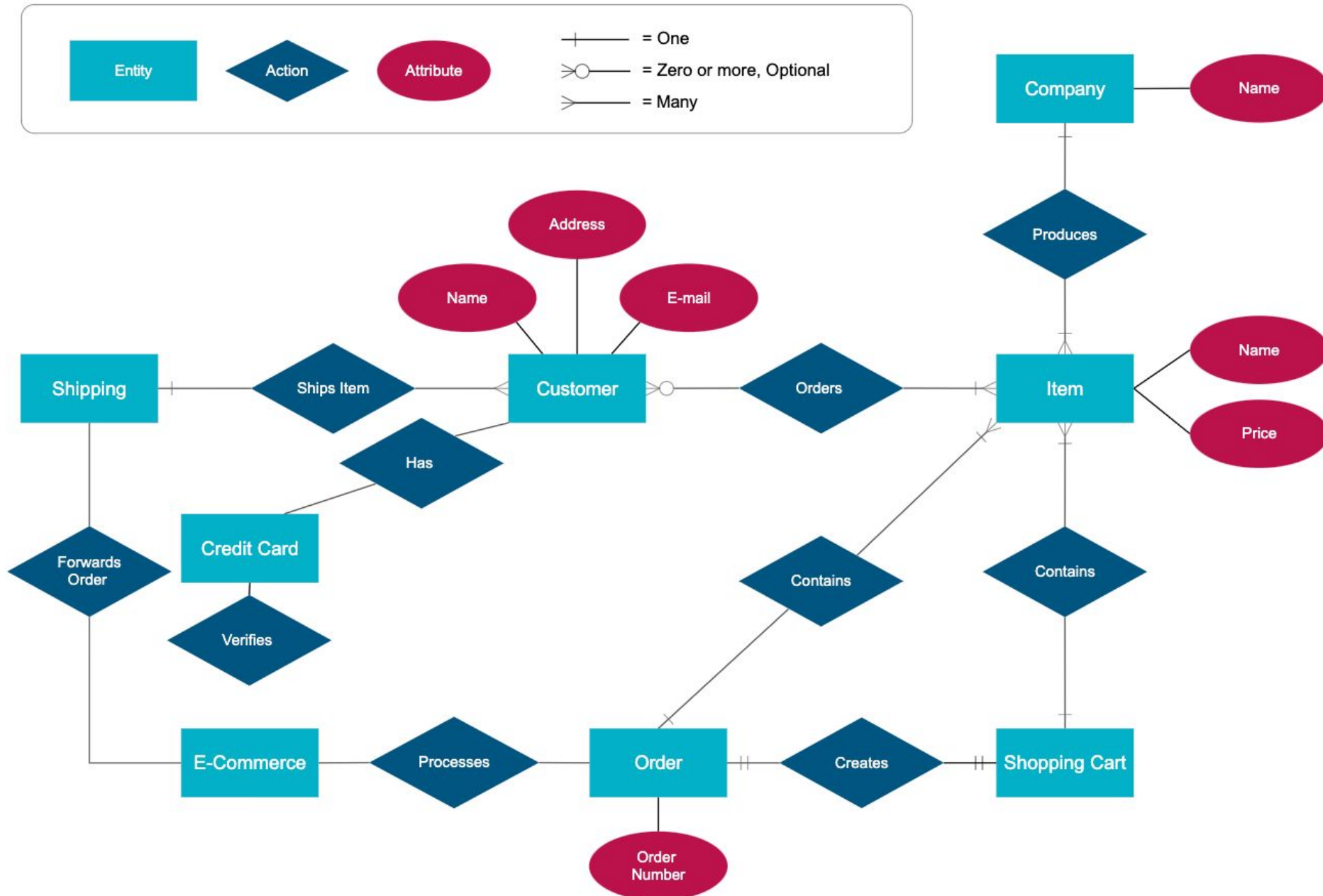
## ? Hospital ER Diagram

- ? The hospital's database and its associated fields are depicted in this ER diagram. The nature, relationships, and qualities of various entities are discussed here. Patients, Doctors, and Tests are the three major entities present in this diagram. This diagram is beneficial to the clinic's productivity and patient retention rates.

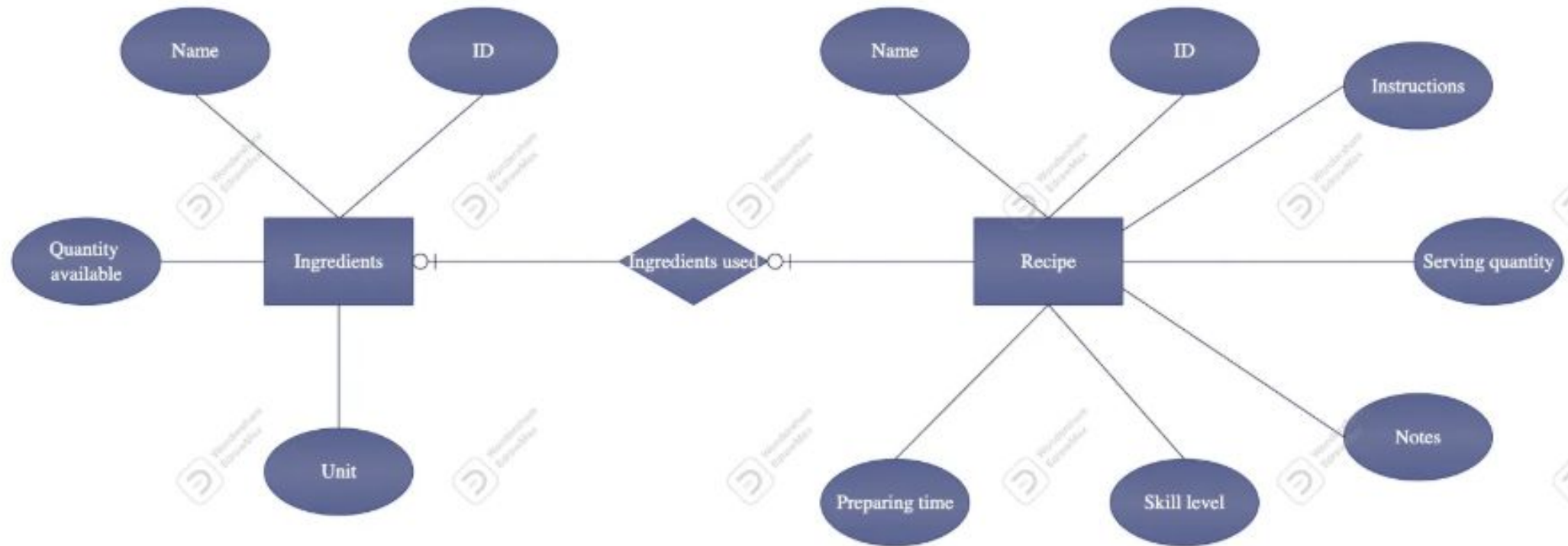
# ER diagram of Hospital



# Entity Relationship Diagram - Internet Sales Model



## ER diagram of Recipe database



# Class Activity-ERD

- ? United Parcel Service (UPS) prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items have attributes such as item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers have attributes such as type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events have attributes such as a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute.